

## Section 1.2 Exercises

For exercises 1 - 20, find the domain of each function using interval notation.

1.  $f(x) = -2x(x - 1)(x - 2)$

2.  $f(x) = 5 - 2x^2$

3.  $f(x) = 3\sqrt{x - 2}$

4.  $f(x) = 3 - \sqrt{6 - 2x}$

5.  $f(x) = \sqrt{4 - 3x}$

6.  $f(x) = \sqrt{x^2 + 4}$

7.  $f(x) = \sqrt[3]{1 - 2x}$

8.  $f(x) = \sqrt[3]{x - 1}$

9.  $f(x) = \frac{9}{x-6}$

10.  $f(x) = \frac{3x+1}{4x+2}$

11.  $f(x) = \frac{\sqrt{x+4}}{x-4}$

12.  $f(x) = \frac{x-3}{x^2+9x-22}$

13.  $f(x) = \frac{1}{x^2-x-6}$

14.  $f(x) = \frac{2x^3-250}{x^2-2x-15}$

15.  $\frac{5}{\sqrt{x-3}}$

16.  $\frac{2x+1}{\sqrt{5-x}}$

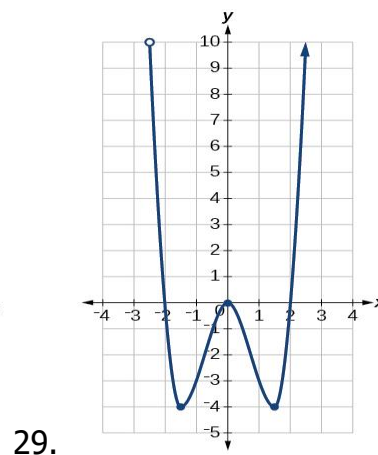
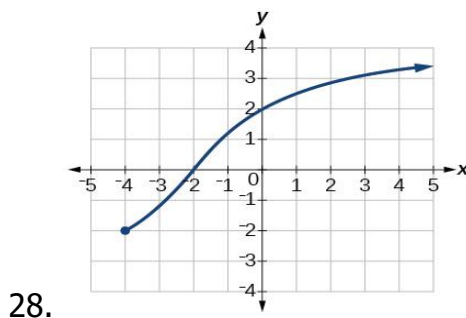
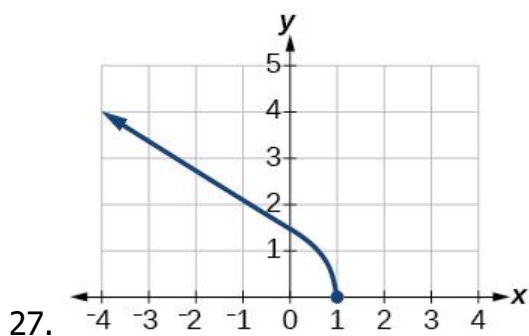
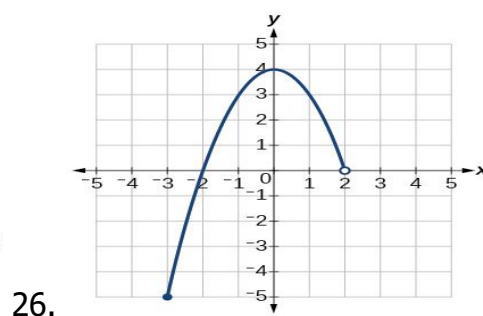
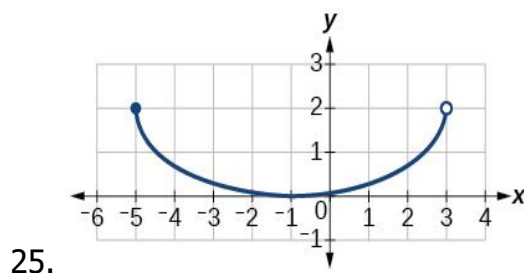
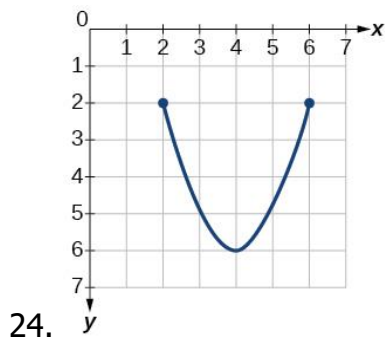
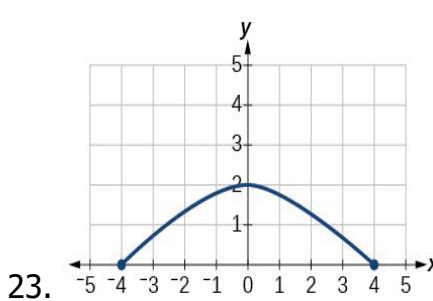
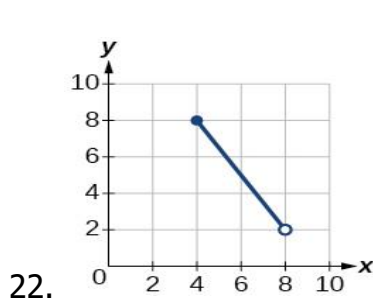
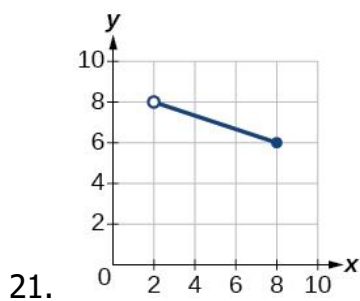
17.  $f(x) = \frac{\sqrt{x-4}}{\sqrt{x-6}}$

18.  $f(x) = \frac{\sqrt{x-6}}{\sqrt{x-4}}$

19.  $f(x) = \frac{x}{x}$

20.  $f(x) = \frac{x^2-9x}{x^2-81}$

For the exercises 21 - 29, write the domain and range of each function using interval notation.



For exercises 30 - 37, sketch a graph of the piecewise function. Write the domain in interval notation.

30. 
$$f(x) = \begin{cases} x + 1 & \text{if } x < -2 \\ -2x - 3 & \text{if } x \geq -2 \end{cases}$$

31. 
$$f(x) = \begin{cases} 2x - 1 & \text{if } x < 1 \\ 1 + x & \text{if } x \geq 1 \end{cases}$$

32. 
$$f(x) = \begin{cases} x + 1 & \text{if } x < 0 \\ x - 1 & \text{if } x > 0 \end{cases}$$

33. 
$$f(x) = \begin{cases} 3 & \text{if } x < 0 \\ \sqrt{x} & \text{if } x \geq 0 \end{cases}$$

$$34. f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ 1 - x & \text{if } x > 0 \end{cases}$$

$$35. f(x) = \begin{cases} x^2 & \text{if } x < 0 \\ x + 2 & \text{if } x \geq 0 \end{cases}$$

$$36. f(x) = \begin{cases} x + 1 & \text{if } x < 1 \\ x^3 & \text{if } x \geq 1 \end{cases}$$

$$37. f(x) = \begin{cases} |x| & \text{if } x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

For exercises 38 - 40, given each function  $f$ , evaluate  $f(-3)$ ,  $f(-2)$ ,  $f(-1)$ , and  $f(0)$ .

$$38. f(x) = \begin{cases} x + 1 & \text{if } x < -2 \\ -2x - 3 & \text{if } x \geq -2 \end{cases}$$

$$39. f(x) = \begin{cases} 1 & \text{if } x \leq -3 \\ 0 & \text{if } x > -3 \end{cases}$$

$$40. f(x) = \begin{cases} -2x^2 + 3 & \text{if } x \leq -1 \\ 5x - 7 & \text{if } x > -1 \end{cases}$$

For exercises 41 - 43, given each function  $f$ , evaluate  $f(-1)$ ,  $f(0)$ ,  $f(2)$ , and  $f(4)$ .

$$41. f(x) = \begin{cases} 7x + 3 & \text{if } x < 0 \\ 7x + 6 & \text{if } x \geq 0 \end{cases}$$

$$42. f(x) = \begin{cases} x^2 - 2 & \text{if } x < 2 \\ 4 + |x - 5| & \text{if } x \geq 2 \end{cases}$$

$$43. f(x) = \begin{cases} 5x & \text{if } x < 0 \\ 3 & \text{if } 0 \leq x \leq 3 \\ x^2 & \text{if } x > 3 \end{cases}$$

For exercises 44 - 46, write the domain for the piecewise function in interval notation.

$$44. f(x) = \begin{cases} x + 1 & \text{if } x < -2 \\ -2x - 3 & \text{if } x \geq -2 \end{cases}$$

$$45. f(x) = \begin{cases} x^2 - 2 & \text{if } x < 1 \\ -x^2 + 2 & \text{if } x > 1 \end{cases}$$

$$46. f(x) = \begin{cases} 2x - 3 & \text{if } x < 0 \\ -3x^2 & \text{if } x \geq 2 \end{cases}$$